WEEK 2 — PROMPT EXPERIMENTATION REPORT

Research Topic: Climate Change

Research Task: Explain the major impacts of climate change on humans and the environment.

Team: RIT 1409 PE TEAM 5

AI Prompt Engineering Research & Integration Internship

TABLE OF CONTENTS

1. Introduction
2. Purpose of the Experiment
3. Team Contributions

4. Deliverable 1 — Prompt Experimentation
   4.1 Research Topic and Task
   4.2 Prompt 1 — Basic/Simple Prompt (Miriam)
   4.3 Prompt 2 — Structured RCTFC Prompt (Roseline)
   4.4 Prompt 3 — Detailed/Controlled RCTFC Prompt (Chiamaka)
   4.5 Comparison of Prompt Variations

5. Deliverable 2 — Output Evaluation Matrix
   5.1 Scoring Methodology
   5.2 Evaluation Matrix
   5.3 Evaluation Justifications
   5.4 Overall Comparison

6. Deliverable 3 — Prompt Optimization Recommendations
   6.1 What Varied Across the Three Prompts
   6.2 What Improved the Responses
   6.3 What None of the Prompts Fixed
   6.4 Recommended Prompt Design Practices

6.5 Future Improvements

7. Overall Findings

8. Conclusion

9. References

1. INTRODUCTION

Prompt design significantly influences the quality, structure, and reliability of AI-generated responses. Yet the relationship between prompt complexity and output quality is not straightforward. This report documents Week 2 of the RIT 1409 PE Team 5 internship, in which three progressively refined prompt variations were tested against the same research task to examine how prompt design decisions affect AI output.

The research task was identical across all three tests: "Explain the major impacts of climate change on humans and the environment." The three prompts tested ranged from a simple one-line request to increasingly structured and controlled variations incorporating the RCTFC framework and explicit anti-fabrication constraints.

2. PURPOSE OF THE EXPERIMENT

The purpose of this experiment was to examine how different prompt designs influence the quality, structure, completeness, relevance, and reliability of AI-generated responses to the same climate-change research task. Three prompt variations were tested: a basic prompt, a structured RCTFC prompt, and a more detailed controlled RCTFC prompt. The experiment aimed to identify which design elements have the most measurable effect on output quality and to develop evidence-based recommendations for prompt design in future research.

3. TEAM CONTRIBUTIONS

The following table documents Week 2 contributions by team member:

Team Member	Contribution
Muqadas Ali (Team Lead)	Team coordination, task distribution, compilation and final submission
Asare Amma Miriam	Prompt 1: Basic/Simple Prompt
Ayansiji Roseline Nike	Prompt 2: Structured RCTFC Prompt
Chiamaka Chinaza	Prompt 3: Detailed/Controlled RCTFC Prompt
Hassan Ridwan	Deliverable 2: Output Evaluation Matrix; Deliverable 3: Prompt Optimization Recommendations
Fizza Mubeen	No Week 2 contribution submitted
Joseph Udeme Nseabsi	No Week 2 contribution submitted
Dhayamtiran S	No Week 2 contribution submitted
Jerry	No Week 2 contribution submitted

4. DELIVERABLE 1 — PROMPT EXPERIMENTATION

4.1 Research Topic and Task

Research Topic: Climate Change

Research Task: Explain the major impacts of climate change on humans and the environment.

4.2 Prompt 1 — Basic/Simple Prompt

Prepared by: Asare Amma Miriam

Prompt Version: Basic/Simple

This is a single-line prompt with no role specification, no defined format, and no constraints.

Full Prompt Used:

"Explain the major impacts of climate change on humans and the environment."

AI-Generated Output:

Climate change acts as a global risk multiplier, reshaping physical systems and altering human societies.

Impacts on the Environment

1. Shifts in Hydrological Cycles & Extreme Weather

Warmer Atmospheres Hold More Water: Evaporation accelerates, causing prolonged, severe droughts in arid regions alongside intense, hyper-concentrated rainfall in others.

Severe Storms: Elevated ocean surface temperatures fuel stronger tropical cyclones, typhoons, and hurricanes with higher rainfall totals and peak wind speeds.

2. Melting Cryosphere & Sea-Level Rise

Ice Loss: Polar ice sheets (Greenland and Antarctica) and mountain glaciers are shrinking rapidly.

Thermal Expansion & Inundation: As ocean water warms, it expands. Combined with glacial melt, global sea levels are rising, causing coastal erosion, saltwater intrusion into freshwater aquifers, and permanent loss of low-lying land.

3. Ecosystem Degradation & Ocean Acidification

Biodiversity Loss: Unprecedented rates of temperature shift force species to migrate poleward or to higher altitudes. Species unable to adapt face extinction.

Marine Disruptions: Oceans absorb roughly 30% of emitted CO_2, forming carbonic acid. This ocean acidification deprives shell-building organisms of carbonate ions, precipitating mass coral bleaching events and disrupting marine food webs.

Impacts on Human Systems

1. Food & Water Insecurity

Agricultural Stress: Shifts in rainfall patterns, heatwaves, and desertification reduce crop yields for staples like wheat, maize, and rice in tropical and subtropical zones.
Freshwater Scarcity: Glacial retreat reduces seasonal runoff for rivers that supply fresh water to hundreds of millions of people, while droughts strain municipal reservoirs.

2. Health Risks
Extreme Heat: Heatwaves lead to higher rates of heat exhaustion, cardiovascular failure, and stroke.
Vector-Borne Diseases: Warmer climates expand the geographic range and active seasons for vectors like mosquitoes and ticks, spreading diseases like malaria, dengue fever, and Lyme disease into higher latitudes and altitudes.

3. Economic Loss & Climate Displacement
Infrastructure Damage: Rising seas and severe storms degrade coastal cities, energy grids, supply chains, and transport networks.
Displacement: Destruction of livelihoods, persistent crop failures, and flooding create millions of "climate refugees" or internally displaced persons, heightening geopolitical tensions over resources.

Observation (Miriam):
What is Good:
• Well-structured with clear headers and numbered subsections
• Much easier to scan and follow
• Includes a specific statistic (oceans absorb 30% of emitted CO_2)

What is Missing:
• No defined audience or purpose
• No source citations for the figures
• No regional specification

4.3 Prompt 2 — Structured RCTFC Prompt

Prepared by: Ayansiji Roseline Nike

Prompt Version: Structured RCTFC

RCTFC Framework:
The RCTFC (Role–Context–Task–Format–Constraints) framework is used to structure the prompt:

Component	Description
Role	Environmental researcher
Context	The research focuses on the effects of climate change on humans and the environment.
Task	Analyze the major environmental, social, economic, and health impacts of climate change.
Format	Use clear headings and concise explanations.

Constraints	Use evidence-based information and avoid unsupported claims.

Full Prompt Used:

"You are an environmental researcher. The research focuses on the effects of climate change on humans and the environment. Analyze the major environmental, social, economic, and health impacts of climate change. Use clear headings and concise explanations. Use evidence-based information and avoid unsupported claims."

AI-Generated Output:

Major Impacts of Climate Change on Humans and the Environment

Climate change affects ecosystems, biodiversity, water resources, food systems, and human societies. Its effects occur through rising temperatures, changing rainfall patterns, sea-level rise, and climate-related hazards.

Environmental Impacts

Climate change affects ecosystems, biodiversity, water resources, and food systems. Rising temperatures and changing rainfall patterns can increase droughts, floods, and other extreme weather events. These changes can damage ecosystems and reduce the ability of some species to survive or adapt.

Social Impacts

Climate change can affect homes, livelihoods, access to water and food, and living conditions. Floods, droughts, storms, and other climate-related hazards can cause displacement and place pressure on communities and essential services.

Economic Impacts

Climate change can cause economic losses by affecting agriculture, fisheries, tourism, infrastructure, and people's ability to work. Extreme weather events can damage infrastructure and disrupt economic activities.

Health Impacts

Climate change affects human health through extreme heat, floods, storms, food insecurity, and changes in some infectious and vector-borne diseases. Climate-related disasters can also affect mental health and disrupt health services.

Conclusion

Climate change has interconnected effects on the environment and human societies. Its impacts can be seen in ecosystems, health, livelihoods, infrastructure, food security, and water security. The severity varies across regions and communities.

Observation (Roseline):

The structured prompt produced a more organized and focused response because it clearly defined the role, context, task, format, and constraints. The answer addressed environmental, social, economic, and health

impacts under separate headings and followed the instruction to use evidence-based information. However, it could be improved further by requesting specific sources, dates, examples, or statistics for each impact.

Evidence Sources Cited:

• Intergovernmental Panel on Climate Change (IPCC), Sixth Assessment Report.

• World Health Organization (WHO), Climate Change and Health.

4.4 Prompt 3 — Detailed/Controlled RCTFC Prompt

Prepared by: Chiamaka Chinaza

Prompt Version: Detailed/Controlled RCTFC

Full Prompt Used:

"Act as an environmental research analyst, examine how climate change affects humans and the environment. Analyze the environmental, social, economic, and health impacts of climate change. Organize the answer into Introduction, Key Findings, Impacts, Limitations, and Conclusion. Use credible evidence/sources, do not invent statistics or sources, distinguish facts from uncertainty and avoid unsupported conclusions."

AI-Generated Output:

Introduction

Climate change, driven primarily by human activities—most notably the combustion of fossil fuels, land-use changes, and industrial emissions—presents a system-wide threat to both natural ecosystems and human civilization. Global mean surface temperatures have increased by approximately 1.1°C to 1.2°C above pre-industrial levels (1850–1900 baseline). This shift alters hydrological cycles, shifts atmospheric patterns, and increases the frequency and severity of extreme weather occurrences.

Addressing these systemic risks requires synthesizing findings across environmental, social, economic, and health domains while maintaining a clear distinction between empirically observed facts and projection-based uncertainties.

Key Findings

Human Attribution: According to the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6), it is unequivocal that human influence has warmed the atmosphere, ocean, and land.

Current Trajectory: Greenhouse gas (GHG) concentrations remain at record highs; atmospheric carbon dioxide (CO_2) exceeds 50% above pre-industrial levels. Without rapid decarbonization, warming is projected to surpass 1.5°C within the mid-21st century.

Vulnerability & Exposure: Approximately 3.3 to 3.6 billion people live in contexts highly vulnerable to climate change. Vulnerability is disproportionately concentrated in developing nations, coastal environments, and low-income populations.

Irreversibility: Certain changes, such as deep-ocean warming, ice-sheet loss, and sea-level rise, are locked in and irreversible on multi-century to millennial timescales.

Impacts

1. Environmental Impacts

Ecosystem Integrity & Species Shifts: Approximately half of assessed species globally have shifted their geographic ranges poleward or to higher elevations. Ocean acidification and marine heatwaves have driven widespread coral bleaching and severe habitat degradation.

Hydrological Dynamics: Warmer air temperatures accelerate evaporation and hold more moisture, compounding both intense drought conditions in arid regions and heavy precipitation events elsewhere.

Cryosphere & Oceans: Glaciers continue to retreat globally, and Arctic sea ice is diminishing at unprecedented rates. Global average sea level has risen by approximately 20–24 cm since 1880, escalating coastal erosion and tidal flooding.

2. Social Impacts

Displacement & Migration: Flooding, shoreline erosion, land degradation, and resource scarcity drive localized displacement and forced internal migration, particularly in small island developing states (SIDS) and river deltas.

Food Security: Shifts in precipitation and extreme heat undermine crop yields (e.g., staple grains) and marine fisheries productivity, threatening food access in climate-sensitive regions.

Equity & Justice: Low-emission societies often suffer the most severe impacts, worsening pre-existing social inequalities, gender disparities, and regional instabilities.

3. Economic Impacts

Infrastructure Loss: Extreme weather events—such as intense hurricanes, wildfires, and riverine floods—cause billions of dollars in damage to transport, energy, and residential infrastructure.

Labor Productivity: High ambient temperatures reduce human physical capacity, particularly in outdoor sectors like construction and agriculture, lowering labor output.

Financial Risk: Macroeconomic risks include growing insurance gaps, depreciated coastal real estate assets, and financial system volatility associated with supply-chain disruptions.

4. Health Impacts

Direct Thermal & Extreme Events: Rising frequency of heatwaves causes elevated mortality from heatstroke and cardiovascular stress.

Vector-Borne & Infectious Diseases: Shifting temperature and rainfall patterns expand the geographical vectors for diseases such as malaria, dengue fever, and waterborne pathogens.

Air Quality & Mental Health: Wildfire smoke and elevated ozone levels exacerbate direct respiratory conditions. Additionally, repeated exposure to weather disasters increases trauma, anxiety, and post-traumatic stress disorders.

Limitations

Climate Modeling & Regional Resolution: While global climate models accurately capture broad temperature trends, downscaling models to predict precise local-level weather shifts, localized rainfall changes, and urban microclimates carries inherent uncertainty.

Feedback Loops & Tipping Points: The exact temperature thresholds required to trigger self-reinforcing climate feedback loops—such as massive permafrost thawing or the collapse of the Atlantic Meridional Overturning Circulation (AMOC)—remain subject to ongoing research.

Socioeconomic Projections: Long-term risk projections rely on Shared Socioeconomic Pathways (SSPs). Future impacts depend heavily on human policy choices, technological development, and emissions trajectories that cannot be predicted with total certainty.

Attribution Complexities: Distinguishing direct climate-driven impacts from non-climate factors (e.g., local deforestation, land development, or poverty dynamics) requires multi-variable analysis and localized data, which is often limited in low-income regions.

Conclusion

The scientific consensus established by global monitoring networks and peer-reviewed syntheses demonstrates that climate change poses existential risks across environmental, economic, social, and human health domains. Current global mitigation and adaptation efforts fall short of the targets required to limit global warming to 1.5°C. Addressing these challenges requires rapid, systemic reductions in global greenhouse gas emissions paired with targeted, equity-focused adaptation strategies to safeguard human well-being and maintain environmental stability.

Observation (Chiamaka):

Credible sources and evidence from the web were provided for each of the responses given. Significant data points were summarized. A high-level summary was given under the conclusion.

4.5 Comparison of Prompt Variations

The three prompts demonstrate a progression from minimal structure to explicitly controlled RCTFC design:

Prompt 1 (Miriam) was a single-line, unstructured request with no role, no defined categories, and no format instruction. The AI still produced a reasonably organized response, suggesting some default organizational capability.

Prompt 2 (Roseline) added RCTFC structure, assigning a role, context, a defined task with four named impact categories (environmental, social, economic, and health), a format instruction, and a constraint to use evidence-

based information. This produced a more uniformly organized response that followed the four categories with consistent section headings.

Prompt 3 (Chiamaka) kept the role and categories but went further, specifying an exact output structure (Introduction, Key Findings, Impacts, Limitations, Conclusion) and adding an explicit anti-fabrication constraint (do not invent statistics or sources, distinguish facts from uncertainty). This produced the most comprehensive response, with additional sections not requested in the simpler prompts.

The key observation is that each step added a different design element. Prompt 2 introduced role and category naming. Prompt 3 introduced exact output structure and explicit verification constraints. The evaluation matrix in Deliverable 2 helps identify which of these additions had the most measurable effect.

5. DELIVERABLE 2 — OUTPUT EVALUATION MATRIX

Prepared by: Hassan Ridwan

5.1 Scoring Methodology

All three outputs were evaluated using a consistent scoring framework:

Scoring Key:
5 = Excellent
4 = Strong
3 = Adequate
2 = Weak
1 = Very weak

Special Note on Hallucination Risk and Bias Risk:
These two criteria are scored in the opposite direction:

5 = Very low risk
1 = Very high risk

Criteria Evaluated:

1. Accuracy: Factual correctness of claims, including statistics and figures
2. Clarity: Readability and comprehensibility of the output
3. Completeness: Extent to which the response addresses the full scope of the research task
4. Relevance: Degree to which content stays focused on the research question
5. Structure: Organization and logical flow of the response
6. Hallucination Risk: Likelihood that the response contains fabricated, unsourced, or verifiably false information
7. Bias Risk: Likelihood that the response presents skewed, one-sided, or undisclosed normative framing

5.2 Evaluation Matrix

Criterion	Prompt 1 (Basic)	Prompt 2 (RCTFC)	Prompt 3 (Detailed)
Accuracy	3	3	4
Clarity	4	4	4
Completeness	3	2	5
Relevance	4	4	5
Structure	3	3	5
Hallucination Risk	3	3	4
Bias Risk	4	4	4

5.3 Evaluation Justifications

Accuracy

Prompt 1 (Score 3): The output includes the statistic that "oceans absorb roughly 30% of emitted CO_2," which checks out against NOAA data. However, this figure is not sourced within the response itself, creating verification difficulty.

Prompt 2 (Score 3): The statements are broadly consistent with established climate science, but they are so general and lack specific numbers that they provide little concrete evidence to assess or verify.

Prompt 3 (Score 4): The output includes several major figures that hold up against verification: the 1.1°C to 1.2°C warming figure, the 3.3 to 3.6 billion vulnerable people figure, and the "unequivocal" human-attribution finding from IPCC AR6 are accurate. However, two quantitative claims require qualification. The claim that sea level has risen by "approximately 20–24 cm since 1880" does not precisely match AR6's 20 cm between 1901 and 2018. The claim that CO_2 "exceeds 50% above pre-industrial levels" runs ahead of the roughly 47% increase AR6 actually reports for the 1750–2019 period. These two discrepancies prevent a perfect 5 score.

Clarity

All three outputs read clearly. Prompt 1 uses headers and subsections for easy scanning. Prompt 2 is short and plain, avoiding jargon. Prompt 3 is well organized with structured sections but includes technical terms (e.g., "SSPs," "AMOC") that require some climate science background to follow fully. All receive a 4.

Completeness

Prompt 1 (Score 3): Covers environmental and human impacts adequately but without the breadth of later responses.

Prompt 2 (Score 2): Hits all four requested categories (environmental, social, economic, health) with clear headings but provides barely one paragraph per category, making the response thin.

Prompt 3 (Score 5): Covers the four impact categories plus an Introduction, Key Findings section, a dedicated Limitations section, and a Conclusion. This is the fullest and most comprehensive response.

Relevance

Prompt 1 (Score 4): Stays on task throughout.

Prompt 2 (Score 4): Stays on task, addressing the four named impact categories.

Prompt 3 (Score 5): Fully on task. While it includes an "Equity & Justice" point within Social Impacts (rather than as a separate section), this integration keeps the response tightly focused rather than allowing scope creep.

Structure

Prompt 1 (Score 3): The response improvises a reasonable structure without being explicitly directed to do so. Sections are organized but not structured to a formal template.

Prompt 2 (Score 3): Follows the instruction to use clear headings and produces four headings (one per impact category) plus a conclusion, but the structure is minimal.

Prompt 3 (Score 5): The exact output structure matches what the prompt requested (Introduction, Key Findings, Impacts, Limitations, Conclusion), demonstrating precise compliance with the structural specification.

Hallucination Risk

Prompt 1 (Score 3): Includes a specific, verifiable statistic (the 30% ocean CO_2 figure) without a source, creating an identifiable verification risk. The specificity of the claim increases the risk.

Prompt 2 (Score 3): Makes mostly broad, general claims about climate change, which reduces the number of specific items that could be verifiably fabricated. However, the generality also makes it difficult to assess the evidentiary basis either way.

Prompt 3 (Score 4): The anti-fabrication instruction ("do not invent statistics or sources, distinguish facts from uncertainty") appears to have reduced hallucination risk. Most numbers held up against verification. However, the two unqualified figures (sea-level rise and CO_2 percentage) show that the instruction reduced but did not eliminate hallucination risk.

Bias Risk

Prompt 1 (Score 4): Uses neutral scientific framing with no strong normative language or ideological positioning.

Prompt 2 (Score 4): Also uses neutral language and avoids ideological framing, though its broad and thin treatment leaves limited basis to judge whether all relevant perspectives or uncertainties were included.

Prompt 3 (Score 4): Uses neutral scientific framing and appropriately includes an "Equity & Justice" point that acknowledges distributional impacts without imposing ideology. The response maintains objectivity while addressing legitimate distributional aspects of climate impacts.

5.4 Overall Comparison

Based on the evaluation matrix:

• Prompt 3 (Detailed/Controlled RCTFC) achieved the strongest overall performance, particularly on Completeness, Relevance, and Structure.

• Prompt 2 (Structured RCTFC) produced a more organized response than Prompt 1 but scored lower on Completeness because each category received minimal elaboration.
• Prompt 1 (Basic) demonstrated that even without explicit instruction, the AI improvised reasonable organization and included specific, relevant content, though it lacked structure and source attribution.

Important caveat: Prompt 3's higher scores on structure and completeness do not guarantee factual accuracy. The two quantitative discrepancies in Prompt 3's output (regarding sea-level rise and CO_2 percentage) demonstrate that explicit output structure and anti-fabrication instructions improve organization and reduce—but do not eliminate—verification risk.

6. DELIVERABLE 3 — PROMPT OPTIMIZATION RECOMMENDATIONS

Prepared by: Hassan Ridwan

6.1 What Varied Across the Three Prompts

Prompt 1 (Miriam) was a single-line, unstructured request with no role, no named categories, and no format instruction.

Prompt 2 (Roseline) added an RCTFC structure, assigning a role ("environmental researcher"), context, a defined task specifying four named impact categories (environmental, social, economic, and health), a format instruction ("use clear headings and concise explanations"), and a constraint ("use evidence-based information and avoid unsupported claims").

Prompt 3 (Chiamaka) kept the role and categories but went further. It specified an exact output structure (Introduction, Key Findings, Impacts, Limitations, Conclusion) and added an explicit anti-fabrication constraint ("do not invent statistics or sources, distinguish facts from uncertainty and avoid unsupported conclusions").

Each step added a different type of design element. The evaluation matrix in Deliverable 2 helps identify which additions had the clearest effect.

6.2 What Actually Improved the Responses

Explicit Output Structure Had the Clearest Effect
Prompt 3's specification of exact output sections (Introduction, Key Findings, Impacts, Limitations, Conclusion) correlated with higher scores on Completeness and Structure. Prompts 1 and 2 were not given specific sections to follow. Prompt 1 still improvised a reasonable structure, while Prompt 2 followed only the looser instruction to use "clear headings" and produced a thinner response. The lesson is not that more detail is automatically better, but that naming the sections you want makes the expected output clearer and more complete.

Anti-Fabrication Instructions Reduced Risk but Did Not Eliminate It

Prompt 3 was explicitly told not to invent statistics or sources and to distinguish facts from uncertainty. Most of its numbers held up against AR6 verification. However, two required correction (the sea-level rise and CO_2 percentage figures). Telling a model not to hallucinate reduces risk, but it does not guarantee accuracy. The output still needs fact-checking rather than being treated as reliable simply because the prompt included an anti-hallucination instruction.

Role-Framing on Its Own Did Not Have a Clear Isolated Effect
Prompt 2 assigned a role ("environmental researcher") and produced a more organized response than Prompt 1. However, that improvement is more closely associated with the four named categories and the formatting instruction than with the role itself. Prompt 2's content remained generic, and in some areas (completeness), was even thinner than Prompt 1's.

Naming Categories Directly Improves Relevance and Completeness When Paired With Depth Requirements
Prompt 2 asked for four categories and received four thin paragraphs—one per category. Prompt 3 asked for the same four categories plus a defined structure and stronger content controls, and received a fuller answer. Category-naming alone is insufficient; it needs a companion instruction pushing for elaboration.

6.3 What None of the Prompts Fixed

None of the three prompts required citations tied to individual claims. Prompt 2 listed two sources (IPCC AR6, WHO) at the end, disconnected from specific sentences. Prompt 3 said to use "credible evidence/sources" in general terms and still produced two quantitative claims that required correction. Neither approach makes it easy for a reader to check a specific number against its source without doing the verification manually after receiving the output.

6.4 Recommended Prompt Design Practices

1. Name the Exact Output Sections You Want
Rather than simply asking for "clear headings," specify the exact sections desired. A defined structure produces a more complete and better-organized answer than a loose formatting instruction.

2. Pair Anti-Hallucination Instructions With Verification
Rather than treating anti-hallucination instructions as a substitute for fact-checking, treat them as a floor. Instructions reduce risk but do not guarantee accuracy, especially for numeric claims. Always plan for a verification step.

3. Require Inline Attribution for Specific Claims
Instead of a general "cite your sources" instruction, require specific attribution next to each claim. Something like "next to any number, name the source and year" would make quantitative claims easier to verify and could help catch mismatches before the final answer is delivered.

4. Combine Category-Naming With Depth Requirements
Pair a list of categories with a depth requirement such as "give at least two specific, evidenced points per category." This avoids thin, generic output that technically addresses all categories but provides minimal elaboration.

5. Do Not Assume More Structure Equals More Accuracy
Prompt 3 was the most detailed and best-organized response and still contained two quantitative claims requiring correction. Detail and organization improve Structure and Completeness but do not guarantee Accuracy. These are separate concerns requiring separate approaches.

6.5 Future Improvements

Future prompt designs should incorporate the recommended practices above, particularly inline attribution requirements and explicit verification steps. Additionally, prompts could benefit from specifying what types of claims require sources (e.g., "all quantitative claims must include a source and year") rather than asking for general citation. Testing these refined approaches on additional research topics would strengthen the evidence base for prompt design best practices.

7. OVERALL FINDINGS

The three-prompt experiment reveals a nuanced relationship between prompt design and output quality:

Prompt 1 (Basic) demonstrated that even without explicit instruction, AI can produce useful, reasonably organized responses. The output covered both environmental and human impacts with relevant examples and included a specific, verifiable statistic.

Prompt 2 (Structured RCTFC) introduced RCTFC framework elements and produced a more uniformly organized response with clearly delineated categories. However, the completeness score declined relative to Prompt 1 because each category received minimal elaboration despite the request for structured content.

Prompt 3 (Detailed/Controlled RCTFC) added explicit output structure and anti-fabrication constraints, producing the most complete and best-organized response according to the evaluation. However, completion and organization did not guarantee factual accuracy: two quantitative claims required verification and correction against AR6.

Key finding: Greater prompt detail improved organization, completeness, and relevance. Explicit anti-fabrication instructions reduced (but did not eliminate) hallucination risk. However, factual accuracy remained dependent on post-generation verification, particularly for statistics and specific quantitative claims.

The experiment demonstrates that prompt design is a multi-dimensional problem. Clear output structure, category specification, and explicit constraints each affect different aspects of response quality. Optimization requires tailoring prompt elements to the specific quality dimensions that matter most for the task at hand.

8. CONCLUSION

This Week 2 experiment tested whether progressively refined prompt designs improve AI-generated responses to a climate-change research task. The results demonstrate that prompt design significantly influences output organization, completeness, and structure. Explicit specification of output sections, named categories, anti-fabrication constraints, and depth requirements each contributed measurably to response quality.

However, the most important finding is that structural and organizational improvements do not automatically translate to factual accuracy. The most detailed and well-organized response (Prompt 3) still contained two quantitative claims requiring correction. This underscores a critical lesson: prompt design can improve how information is presented and organized, but it cannot replace fact-checking and verification, particularly for numerical claims and specific factual assertions.

For practitioners using AI for research and analysis, the implications are clear: structure your prompts to guide organization and completeness, pair anti-hallucination language with mandatory verification steps, require inline attribution for specific claims, and treat the final output as a draft requiring review rather than as a finished product.

Effective prompt design is a tool for improving communication between human and AI, but it is not a substitute for critical evaluation of AI-generated content.

9. REFERENCES

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End of Week 2 Prompt Experimentation Report

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